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USE master;
GO
-- حذف القاعدة إذا كانت موجودة لضمان نظافة التنفيذ
IF EXISTS (SELECT * FROM sys.databases WHERE name = 'HealthcareDB')
BEGIN
    ALTER DATABASE HealthcareDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE HealthcareDB;
END
GO

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-- 1. إنشاء القاعدة مع المجموعات (Filegroups)
CREATE DATABASE HealthcareDB
ON PRIMARY (
    NAME = 'HealthcareDB_Primary',
    FILENAME = 'D:\SQLData\HealthcareDB_Primary.mdf',
    SIZE = 10MB, FILEGROWTH = 5MB
),
FILEGROUP PatientDoctorFG (
    NAME = 'HealthcareDB_PatientDoctor',
    FILENAME = 'D:\SQLData\HealthcareDB_PatientDoctor.ndf',
    SIZE = 10MB, FILEGROWTH = 5MB
),
FILEGROUP BillingTreatmentFG (
    NAME = 'HealthcareDB_BillingTreatment',
    FILENAME = 'D:\SQLData\HealthcareDB_BillingTreatment.ndf',
    SIZE = 10MB, FILEGROWTH = 5MB
)
LOG ON (
    NAME = 'HealthcareDB_log',
    FILENAME = 'D:\SQLData\HealthcareDB_log.ldf',
    SIZE = 5MB, FILEGROWTH = 5MB
);
GO

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--2 انشاء الجداول
USE HealthcareDB;
GO

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-- جدول المرضى
CREATE TABLE Patients (
    PatientID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL, -- قيد عدم الفارغ
    DOB DATE CHECK (DOB < GETDATE()), -- قيد التحقق من التاريخ
    Gender VARCHAR(10) CHECK (Gender IN ('Male', 'Female')),
    ContactInfo VARCHAR(255) UNIQUE -- قيد التفرد
) ON PatientDoctorFG;

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-- جدول الأطباء
CREATE TABLE Doctors (
    DoctorID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Specialization VARCHAR(100) DEFAULT 'General Medicine', -- قيد القيمة الافتراضية
    ContactInfo VARCHAR(255)
) ON PatientDoctorFG;

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-- جدول المعالجات
CREATE TABLE Treatments (
    TreatmentID INT PRIMARY KEY,
    PatientID INT FOREIGN KEY REFERENCES Patients(PatientID),
    DoctorID INT FOREIGN KEY REFERENCES Doctors(DoctorID),
    Description TEXT NOT NULL,

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    TreatmentDate DATETIME DEFAULT GETDATE()
) ON BillingTreatmentFG;

-- جدول الفواتير
CREATE TABLE Billing (
    BillID INT PRIMARY KEY,
    TreatmentID INT FOREIGN KEY REFERENCES Treatments(TreatmentID),
    Amount DECIMAL(10, 2) CHECK (Amount > 0), -- التأكد من أن المبلغ إيجابي
    Status VARCHAR(20) DEFAULT 'Unpaid'
) ON BillingTreatmentFG;
GO

-- 1. نسخة احتياطية كاملة
BACKUP DATABASE HealthcareDB
TO DISK = 'D:\SQLBackups\HealthcareDB_Full.bak'
WITH FORMAT, MEDIANAME = 'HospitalFullBackup';

-- 2. محاكاة فشل (حذف جدول)
-- DROP TABLE Billing;

-- 3. استعادة قاعدة البيانات
USE master;
RESTORE DATABASE HealthcareDB
FROM DISK = 'D:\SQLBackups\HealthcareDB_Full.bak'
WITH REPLACE;

```